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Nucleoside Diphosphokinases

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I. Introduction

A. HISTORICAL DEVELOPMENT

The term "nucleoside diphosphokinase" (NDP kinase, ATP:nucleoside diphosphate phosphotransferase, EC 2.7.4.6) (1) is used to designate a family of relatively unspecific enzymes that catalyze the transfer of

1. Since NDP kinases are not specific for ATP, a more correct terminology would be nucleoside triphosphate:nucleoside diphosphate phosphotransferase.

the terminal phosphate group of 5'-triphosphate nucleotides to 5'-diphosphate nucleotides (2) by the following general mechanism:



where N_1 and N_2 are purine or pyrimidine ribo- or deoxyribonucleosides. All NDP kinases that have been examined to date function through the formation of enzyme-bound high-energy phosphate intermediates.

The first NDP kinases clearly identified were reported independently in 1953 by Berg and Joklik for yeast (3) and by Krebs and Hems for pigeon breast muscle (4). Over the next few years, various workers demonstrated that the enzyme is ubiquitously distributed in nature and is found in animals, plants, and microorganisms (see Section I,B).

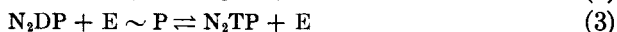
A notable achievement was the crystallization of NDP kinase from brewer's yeast by Ratliff *et al.* (5). Today, the only crystalline preparations of NDP kinase available are those from yeast, although several laboratories (5, 6) have purified NDP kinases from various sources to apparent homogeneity (6-10). The only precise studies of the subunit structure, physicochemical properties, amino acid end groups, and composition of an NDP kinase have been performed with a crystalline yeast preparation which has been found to be a homologous hexamer with a monomer of about 17,000 daltons (11, 12).

A significant development in the understanding of the biochemical behavior of NDP kinases came with the discovery that a purified human erythrocytic enzyme follows a ping-pong reaction sequence and functions through the formation of a stable phosphate to enzyme bond (7, 13, 14). All NDP kinases studied subsequently have been shown to function by

2. Throughout this article it is assumed that actual substrates are Mg-nucleotide complexes, e.g., Mg-ATP.

3. P. Berg and W. K. Joklik, *Nature (London)* **172**, 1008 (1953).
4. H. A. Krebs and R. Hems, *BBA* **12**, 172 (1953).
5. R. L. Ratliff, R. H. Weaver, H. A. Lardy, and S. A. Kuby, *JBC* **239**, 301 (1964).
6. B. Edlund, L. Rask, P. Olsson, O. Wälinder, Ö. Zetterqvist, and L. Engström, *Eur. J. Biochem.* **9**, 451 (1969).
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8. R. P. Agarwal and R. E. Parks, Jr., *JBC* **246**, 2258 (1971).
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10. B. Edlund, *Acta Chem. Scand.* **25**, 1370 (1971).
11. R. H. Yue, R. L. Ratliff, and S. A. Kuby, *Biochemistry* **6**, 2923 (1967).
12. R. Palmieri, R. Yue, H. Jacobs, L. Maland, L. Wu, and S. A. Kuby, *Fed. Proc., Fed. Amer. Soc. Exp. Biol.* **29**, A914 (1970).
13. N. Mourad and R. E. Parks, Jr., *BBRC* **19**, 312 (1965).
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a similar reaction mechanism (8, 9, 15-18). The amino acid phosphorylated, in most instances, has proved to be histidine with the phosphate either in the 1 or the 3 position (6, 9, 10, 19-23). These observations permitted formulation of the following reaction sequence (13):



A recent development of considerable importance to future studies of NDP kinases is the discovery of marked electrophoretic heterogeneity of the enzyme in a number of tissues (22, 24, 25); for example, in human erythrocytes six peaks of NDP kinase activity were demonstrated with isoelectric points ranging from 5.4 to 8.3. The enzymic behavior and molecular properties of these presumed isozymes differ so greatly that questions have been raised concerning the true physiological role of this enzyme (22, 24).

For an excellent and detailed review of studies with NDP kinases prior to 1960, the reader is referred to the chapter by Robert H. Weaver in the second edition of "The Enzymes" (26).

B. DISTRIBUTION

The ubiquitous distribution of NDP kinases is shown in Table I. (27-56). Since the various laboratories have employed different assay

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19. A. W. Norman, R. T. Wedding, and M. K. Black, *BBRC* **20**, 703 (1965).
20. O. Wälinder, *JBC* **243**, 3947 (1968).
21. O. Wälinder, *JBC* **244**, 1065 (1969).
22. Y. C. Cheng, B. Robison, and R. E. Parks, Jr., *Biochemistry* **12**, 5 (1973).
23. O. Wälinder, *Acta Chem. Scand.* **23**, 339 (1969).
24. Y. C. Cheng, R. P. Agarwal, and R. E. Parks, Jr., *Biochemistry* **10**, 2139 (1971).
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26. R. H. Weaver, "The Enzymes," 2nd ed., Vol. 6, p. 151, 1962.
27. R. P. Agarwal, E. M. Scholar, K. C. Agarwal, and R. E. Parks, Jr., *Biochem. Pharmacol.* **20**, 1341 (1971).
28. R. P. Glaze and C. L. Wadkins, *JBC* **242**, 2139 (1967).
29. D. R. Sanadi, D. M. Gibson, P. Ayengar, and M. Jacob, *JBC* **218**, 505 (1956).
30. H. Nakamura and Y. Sugino, *JBC* **241**, 4917 (1966).
31. P. Berg and W. K. Joklik, *JBC* **210**, 657 (1954).

TABLE I
DISTRIBUTION OF NUCLEOSIDE DIPHOSPHOKINASES

Source	Activity ^a (units ^b /g tissue)	Specific activity ^a (units ^b /mg protein)
Mammals		
Human		
Erythrocytes ^c	33.0 (7); 75.0 (8); — ^d ; (20, 24) 76.0 (27)	0.05 (7); 0.11 (8, 27)
Bovine		
Liver	— ^d (20)	
Liver mitochondria	5.4 (28)	0.6 (28)
Heart mitochondria	— ^d (15, 29)	0.225 (15)
Thymus	14.5 (30)	0.12 (30)
Brain (acetone powder)	— ^d (31)	
Rabbit		
Muscle	— ^d (3, 31)	0.1–0.13 (31)
Red bone marrow	— ^d (32)	
Erythrocytes ^c	70.0 (33)	0.11 (33)
Pig		
Kidney	— ^d (34, 35)	
Heart	— ^d (34)	
Rat		
Liver	10.0 (7); 62.0 (22)	0.02 (7); 0.43 (22);
Liver mitochondria—whole	— ^d (36–42); 4.3 (22)	0.017 (43)
Inner membrane [low-speed pellet (2,000–12,000 × <i>g</i> 10 min)]		0.001 (43)
Outer membrane [high-speed pellet (144,000 × <i>g</i> 1 hr)]		0.075 (43)
High-speed supernatant (144,000 × <i>g</i> 1 hr)		0.078 (43)
Liver nuclei	7.44 (22)	
Liver cytosomes and microbodies	3.1 (22)	
Liver endoplasmic reticulum	0.0 (22)	
Liver supernatant	47.0 (22)	
Kidney	27.0 (7); 62.0 (22)	0.10 (7); 0.38 (22)
Heart	25.0 (7); 68.0 (22)	0.08 (7); 0.87 (22)
Spleen	30.0 (7); 63.0 (22)	0.08 (7); 0.16 (22)
Brain	17.0 (7); 64.0 (22)	0.15 (7); 0.84 (22)
Lungs	30.0 (22)	0.24 (22)
Erythrocytes ^c	30.0 (7); 47.0 (22)	0.05 (7); 0.31 (22)
Intestinal mucosa	— ^d (4)	
Baboon		
Erythrocytes ^c	68.0 (33)	0.10 (33)

TABLE I (Continued)

Source	Activity ^a (units ^b /g tissue)	Specific activity ^a (units ^b /mg protein)
Monkey		
Erythrocytes ^c	51.0 (33)	0.09 (33)
Dog		
Erythrocytes ^c	110.0 (33)	0.15 (33)
Cat		
Erythrocytes ^c	90.0 (33)	0.17 (33)
Spleen of leukemic mice	Highly active (44)	
Novikoff hepatoma	— ^d (45)	
Seal, <i>Phoca vitulina</i>	25–45 (25)	
Erythrocytes ^c		
Birds		
Chicken		
Liver mitochondria	— ^d (46)	2.03 (46)
Pigeon		
Breast muscle	— ^d (4)	
Erythrocytes ^c	6.5 (33)	0.01 (33)
Avian myeloblastosis virus	— ^d (47)	
Fishes		
Hagfish, <i>Myxine glutinosa</i>		
Erythrocytes ^c	19.0 (25)	
Dogfish, <i>Squalus acanthias</i>		
Erythrocytes ^c	25.0 (25)	
Eel, <i>Anguilla rostrata</i>		
Erythrocytes ^c	40.0 (25)	
Plants		
Sugar beet leaf	— ^d (48)	
Silver beet leaf	— ^d (49)	
Pea, <i>Pisum sativum</i>		
Seed (dry flour)	100.0 (10); — ^d (49)	0.8 (10)
Root	— ^d (49)	
Shoot	— ^d (49)	
Wheat		
Seed	— ^d (49)	
Shoot	— ^d (49)	
Barley		
Shoot	— ^d (49)	
Broad bean		
Root	— ^d (49)	
Shoot	— ^d (49)	
Sugarcane		
Shoot	— ^d (49)	
Root	— ^d (49)	
Potato tuber	— ^d (49)	

(Continued)

TABLE I (Continued)

Source	Activity ^a (units ^b /g tissue)	Specific activity ^a (units ^b /mg protein)
<i>Impatiens holstii</i> leaves	— ^d (50)	
Jerusalem artichoke, <i>Helianthus tuberosus</i> Mitochondria	— ^d (19)	
Microorganisms		
Brewer's yeast, <i>Saccharomyces calbergensis</i>	— ^d (3, 31); 115.0 ^e (5)	0.88 (31); 0.86 ^f (5)
Baker's yeast	4.4 (6)	0.5 (6)
<i>Bacillus subtilis</i>	— ^d (9, 51)	0.21 (9)
<i>Escherichia coli</i>		
Uninfected	— ^d (47, 52, 53, 53a)	3.3 (53)
Infected	— ^d (47, 52-54)	
<i>Micrococcus luteus</i>	— ^d (47)	
<i>Streptococcus pneumoniae</i>	— ^d (55)	
<i>Schistosoma mansoni</i>	450.0 (56)	

^a The italic numbers in parentheses are reference numbers.

^b Enzymic activities have been converted to a common international unit unless otherwise mentioned. One unit of enzymic activity is that amount of enzyme which catalyzes the reaction of 1 μ mole of nucleoside triphosphate and of nucleoside diphosphate per minute. No attempt has been made to standardize the conditions such as temperature and pH. Hence, strict comparisons are not possible and activities will differ from one report to another.

^c Units/ml packed cells.

^d Activity present but values not reported.

^e Units/g dry yeast.

^f Autolysate.

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34. D. M. Gibson, P. Ayengar, and D. R. Sanadi, *BBA* **21**, 86 (1956).

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36. T. L. Chan, J. W. Greenawalt, and P. L. Pedersen, *J. Cell Biol.* **45**, 291 (1970).

37. R. P. Glaze, *Fed. Proc., Fed. Amer. Soc. Exp. Biol.* **26**, 863 (1967).

38. E. Herbert, V. R. Potter, and Y. Takagi, *JBC* **213**, 923 (1955).

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38b. L. I. Hecht, V. R. Potter, and E. Herbert, *BBA* **15**, 134 (1954).

38c. R. Mantsavinos and E. S. Canellakis, *JBC* **234**, 628 (1959).

39. P. L. Pedersen and C. A. Schnaitman, *JBC* **244**, 5065 (1969).

40. P. L. Pedersen and C. A. Schnaitman, in "Energy Transduction in Respiration and Photosynthesis" (E. Quagliariello *et al.*, eds.), p. 831. Adriatica Editrice, Bari, 1971.

41. C. A. Schnaitman and P. L. Pedersen, *BBRC* **30**, 428 (1968).

conditions, it is not possible to make strict comparisons of the reported values. However, it is apparent that the activity of this enzyme is unusually high in many tissues. For example, in a study of the distribution of NDP kinases in various tissues of the rat, activities ranging from 30 to 68 units/g of tissue were determined (7, 22). Also, the NDP kinase activity in the human erythrocyte, approximately 76 units/ml of packed cells, is one of the highest enzymic activities detected in this cell (27).

II. Molecular Properties

A. OCCURRENCE OF NDP KINASE ISOZYMES

A number of observations suggested that isozymes of NDP kinase occur in various tissues. For example, during attempts at purification of NDP kinases occasional aberrant peaks of activity were observed (6, 7, 28, 30). In a detailed study with the NDP kinase of human erythrocytes, the technique of isoelectric focusing revealed the occurrence of at least six enzymic peaks with isoelectric points ranging from pI 5.4 to 8.3 (Fig. 1). Studies of the kinetic parameters and physico-chemical properties of these activity peaks suggest that they do not represent classic isozymes but rather a family of related enzymes that function through high-energy phosphate intermediates; for example,

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42. C. L. Wadkins and A. L. Lehninger, *Proc. Nat. Acad. Sci. U. S.* **46**, 1576 (1960).
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 51. J. Sedmak, N. Fernald, and R. Ramaley, *ABB* **130**, 488 (1969).
 52. M. P. Argyrakakis, *BBA* **166**, 593 (1968).
 53. L. J. Bello and M. J. Bessman, *BBA* **72**, 647 (1963).
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 54. R. Somerville and G. R. Greenberg, *Fed. Proc., Fed. Amer. Soc. Exp. Biol.* **18**, 327 (1959).
 55. F. S. Markland and C. L. Wadkins, *JBC* **241**, 4124 (1966).
 56. A. W. Senft, G. W. Crabtree, K. C. Agarwal, E. M. Scholar, R. P. Agarwal, and R. E. Parks, Jr., *Biochem. Pharmacol.* **22**, 449 (1973).

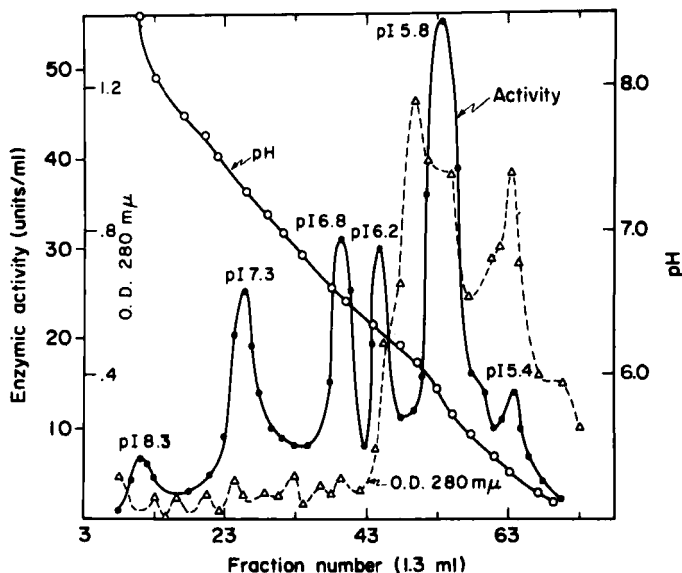


FIG. 1. Electrofocusing profile of human erythrocytic nucleoside diphosphokinase. From Cheng *et al.* (24).

marked differences in the Michaelis constants with various nucleotide substrates were observed. With four of the isozymes, linear Arrhenius plots were seen, whereas biphasic plots were seen with two of the isozymes. Molecular weight studies demonstrated a range of about 80,000–100,000 daltons which indicates that the phenomenon could not have been caused by aggregation (24). More recent studies have demonstrated that the heterogeneity of NDP kinase is not confined to the human but occurs in erythrocytes throughout the animal kingdom (22, 25). Furthermore, a study of the NDP kinases from a variety of tissues of the rat and in the subcellular fractions of rat liver revealed striking heterogeneity of the NDP kinases. However, each organ and subcellular fraction has its own unique NDP kinase isozyme pattern (22).

B. PURIFICATION PROCEDURES

Purification procedures have been developed for NDP kinases from a large number of sources. Detailed procedures have been described for the isolation of crystalline NDP kinases from brewer's and baker's yeast (5, 6). Apparently homogeneous preparations have been obtained from calf liver mitochondria (28), human erythrocytes (7, 8), pea seed (10),

and *Bacillus subtilis* (9). Partial purification has been achieved with NDP kinases from a variety of other sources: beef heart mitochondria (15), calf thymus (30), rabbit muscle (31), and several plant sources (49). The recent discovery of the marked heterogeneity of NDP kinase from several sources (22, 24, 25) suggests that caution should be taken in the selection or development of a purification scheme. The technique of isoelectric focusing has proved especially useful in monitoring for the presence and distribution of isozymes at various stages of purification procedures. As occurred in early studies with erythrocytic NDP kinase (7), the inclusion of a cellulose ion exchange procedure in purification may result in the loss of several isozymes (7, 8, 24). Attention is directed to the potential usefulness of calcium phosphate gel adsorption procedures in the purification of NDP kinases. Where tested, these methods have given virtually quantitative recoveries of enzyme with excellent purification and retention of all the isozymes (8, 22, 24, 27).

C. PHYSICAL PROPERTIES

Several physical properties have been reported for preparations of homogeneous or crystalline NDP kinases as shown in Table II (57, 58). Most NDP kinase preparations studied to date have molecular weights in the range of about 80,000–110,000 daltons. It is of interest that a number of phosphate-binding studies indicate that 3–4 moles of phosphate bind per mole of enzyme, suggesting a tetrameric structure (6, 10, 16). However, the structure of the crystalline yeast enzyme is that of a homologous hexamer with subunits of about 17,000 daltons (12). Excellent agreement was obtained between the molecular weight calculated from the amino acid composition and by physical methods (12). The terminal carboxyl group of the brewer's yeast NDP kinase monomer is a glycine residue, and it was shown that no disulfide bonds were present in the enzyme (12).

D. PHOSPHORYLATED ENZYME

As a consequence of the discovery that the erythrocytic NDP kinase follows a ping-pong reaction sequence, it was predicted that a phosphorylated enzyme intermediate was involved (13). This prediction was verified by incubating [γ - ^{32}P]ATP with purified NDP kinase from human erythrocytes and by separating the phosphorylated enzyme on a

57. P. L. Pedersen, *JBC* **243**, 4305 (1968).

58. P. Andrews, *BJ* **91**, 222 (1964).

TABLE II
SOME PHYSICAL PARAMETERS OF NUCLEOSIDE DIPHOSPHOKINASES FROM VARIOUS SOURCES

Enzyme source	Ref.	Isoelectric point (pI)	Molecular weight	Diffusion coefficient ($d_{20,w} \times 10^7$) ($\text{cm}^2 \text{sec}^{-1}$)	Sedimentation coefficient ($s_{20,w} \times 10^{13}$) (sec)	Apparent specific partial volume $\bar{V}_{\text{app}}$ ($\text{cm}^3 \text{g}^{-1}$)	Frictional ratio (f/f_0)
Human erythrocytes	8, 24	5.4	80,000 ^a	—	—	—	—
		5.8	93,000 ^a	—	—	—	—
		6.3	84,000 ^a	—	—	—	—
		6.8	80,000 ^a	—	—	—	—
		7.3	84,000 ^a	—	—	—	—
		8.3	100,000 ^a	—	—	—	—
Beef heart mitochondria	15	9.5	103,000	—	5.9	—	—
Bovine liver mitochondria	57	—	109,000 ^a	5.49 $\pm$ 0.14	5.64 ^d $\pm$ 0.15	0.77	1.22
			109,000 ^b		5.40 ^e $\pm$ 0.28		
			104,000 ^c				
Pea seed	10	—	70,000	—	—	—	—
Brewer's yeast	11	8.0	102,000	5.63	5.67	0.76	1.19
<i>Bacillus subtilis</i>	9	—	100,000	—	—	—	—
Hagfish erythrocytes	25	5.2; 6.0; 6.7; 7.3	—	—	—	—	—
Eel erythrocytes	25	5.0; 5.8; 6.4; 8.2	—	—	—	—	—
Rat erythrocytes	22	5.3; 5.5; 5.65; 5.8; 6.05; 6.2	—	—	—	—	—
Rat kidney	22	5.3; 5.8; 6.05; 6.2;	—	—	—	—	—

Rat heart	22	6.5 5.3; 5.5; 5.80; 6.05; 6.2; 6.5	—	—	—	—	—
Rat lungs	22	5.3; 5.65; 5.8; 6.05; 6.2; 6.5	—	—	—	—	—
Rat spleen	22	5.3; 5.5; 5.8; 6.05; 6.3; 6.5	—	—	—	—	—
Rat brain	22	5.30; 5.50; 5.65; 5.80; 6.05; 6.20; 6.50	—	—	—	—	—
Rat liver homogenate	22	5.30; 5.50; 5.80; 6.05; 6.20; 6.50	—	—	—	—	—
Nuclei		5.30; 5.50; 5.65; 5.80; 5.90; 6.05; 6.20; 6.50	—	—	—	—	—
Cytosomal fraction		5.30; 5.50; 5.65; 5.80; 5.90; 6.05; 6.20; 6.50	—	—	—	—	—
Mitochondria		5.5; 5.80; 5.95; 6.05; 6.20; 6.50	—	—	—	—	—

(Continued)

TABLE II (Continued)

Enzyme source	Ref.	Isoelectric point (pI)	Molecular weight	Diffusion coefficient ($d_{20,w} \times 10^7$) ($\text{cm}^2 \text{sec}^{-1}$)	Sedimentation coefficient ($s_{20,w} \times 10^{13}$) (sec)	Apparent specific partial volume $\bar{V}_{\text{app}}$ ($\text{cm}^3 \text{g}^{-1}$)	Frictional ratio (f/f_0)
Supernatant		5.30; 5.50; 5.80; 5.90; 6.05; 6.20; 6.50	—	—	—	—	—
Pig kidney	35		21,000 ^c 92,000 ^d 138,000 ^e				

^a By molecular sieving (58).^b From $s_{20,w}$ value of 5.64.^c From $s_{20,w}$ value of 5.40.^d By analytical ultracentrifuge.^e By sucrose gradient.^f Monomer.^g Tetramer.^h Hexamer.

Sephadex column (7, 13, 14). Shortly thereafter these observations were confirmed with an NDP kinase isolated from artichokes (19). In addition, it was demonstrated with this preparation that the high-energy phosphate residue is a phosphorylated histidine (19). Since these original observations, every NDP kinase preparation that has been examined has been shown to function through a high-energy phosphate-enzyme intermediate (6, 9, 10, 16, 18, 20, 21, 23, 35, 51, 57, 59). In a series of investigations with various NDP kinase preparations, Wålander and his colleagues have identified in alkaline digests of phosphorylated NDP kinases 1-phosphohistidine, 3-phosphohistidine, and *N*- ϵ -phospholysine (6, 10, 20). With a NDP kinase purified from bovine liver, 1-phosphohistidine appeared to be the major phosphorylated residue (23). Similarly, with a homogeneous preparation isolated from *B. subtilis*, the principal phosphorylated residue was 1-phosphohistidine (9, 51). In preliminary studies with NDP kinase isozymes from human erythrocytes, both 1-phosphohistidine and 3-phosphohistidine residues were identified (60). A possibility that must be considered in these studies is the migration of the phosphate residue from the 1 to the 3 position of histidine, which is the more stable form, a migration that might occur during the isolation procedure. With most NDP kinases that have been examined, the phosphorylated enzyme intermediates have proved to be alkali stable and acid labile, which is consistent with the formation of phosphate to nitrogen bonds as in phosphorylated histidines (16, 23). However, there have been reports of NDP kinases and of ATP-ADP exchange enzymes that function through alkali-labile and acid-stable phosphorylated enzyme intermediates (35, 59). The acid stability of these phosphorylated enzymes suggests that they may function through carboxyl phosphate residues. The occurrence of apparent NDP kinases with drastically different stability characteristics and phosphorylated amino acid residues suggests that these NDP kinases may serve a function in the cell other than the transfer of phosphate groups among nucleotides.

It has been shown with a number of NDP kinase preparations that more than one mole of phosphate binds per mole of enzyme; for example, with a pea seed NDP kinase, 3.1 moles of phosphate bound per mole of enzyme (10), and with baker's yeast 3.5–3.6 moles of phosphate per mole of enzyme were determined (6). With the human erythrocytic enzyme, approximately 2 moles of phosphate were bound per mole of enzyme (14). Since physical studies have shown that the

59. M. G. Colomb, J. G. Laturaze, and P. V. Vignais, *BBRC* **24**, 909 (1966).
60. Y. C. Cheng, B. Robison, and R. E. Parks, Jr., unpublished data.

brewer's yeast enzyme is a homologous hexamer (12), one should expect that 6 moles of phosphate should bind per mole of enzyme. However, Garces and Cleland have reported four phosphate-binding sites per mole of enzyme (16).

In a series of studies with a rapid mixing technique, the rate of phosphorylation and dephosphorylation was determined with an NDP kinase isolated from bovine liver (18). Here it was shown that the turnover number for the phosphorylation reaction is $2,700 \text{ min}^{-1}$ and for dephosphorylation $24,000 \text{ min}^{-1}$. The turnover number of the overall reaction is about $1,300 \text{ min}^{-1}$, which is strong evidence that the phosphoenzyme is the true intermediate for the NDP kinase reaction (18). It should be noted that the turnover number of crystalline brewer's yeast NDP kinase is very much greater, i.e., $600,000 \text{ min}^{-1}$ or about $150,000 \text{ min}^{-1}$ per active site (16).

III. Catalytic Properties

A. REACTION CATALYZED

The reaction catalyzed by NDP kinases is the readily reversible transfer of the terminal phosphate of a 5'-triphosphate nucleotide to a 5'-diphosphate nucleotide as shown in Eq. (1). A number of studies have reported that the equilibrium constant (K_{eq}) of this reaction is approximately 1.0 (9, 31, 49).

In a detailed study with the crystalline yeast enzyme, Garces and Cleland (16) found that the overall equilibrium constant at pH 8.0 is 1.28. However, the partial equilibrium constant for the phosphorylation of free enzyme by ATP is 0.188, and for dephosphorylation of the phosphorylated enzyme by UDP it is 6.76 as shown in the following equations:



These equilibria favor the free enzyme and suggest that the phosphorylated enzyme has a free energy of hydrolysis about 1.0 kcal more negative than the terminal phosphate bond of ATP (16).

B. SPECIFICITY

All the NDP kinases that have been studied to date are relatively nonspecific and can react with di- and triphosphate ribonucleotides or

deoxyribonucleotides that contain either purine or pyrimidine bases (3, 5, 7, 15, 24, 28, 30, 31, 49). In addition, a number of nucleotides that contain purine or pyrimidine analogs have substrate activity (7). However, despite this relative lack of substrate specificity, one may not conclude that NDP kinases do not discriminate between the various nucleotides since widely different kinetic parameters have been determined (Table III). Furthermore, the NDP kinase isozymes of human erythrocytes vary greatly in their Michaelis constants (Table III) and $V_{\max}$ values with different nucleotides. Although substrate specificities have been described for a number of enzyme preparations from a variety of tissues, little consideration has been given to the isozymic nature of NDP kinases, and it is possible that the substrate specificities described for a specific NDP kinase may, in fact, reflect the presence of mixtures of isozymes that have different reactivities with nucleotide substrates. In view of the marked electrophoretic heterogeneity of NDP kinases in many tissues (22, 24), the whole question of the nucleotide specificity of the various reported preparations should be reopened. It should be noted that monophosphate nucleotides are not capable of serving as substrates for NDP kinases although they may act as weak competitive inhibitors (7).

C. METHODS OF ASSAY

Various procedures have been developed for the measurement of NDP kinase activity. They fall into three major groups: (1) coupled enzymic assays, (2) isotopic assays, and (3) staining procedures for localizing NDP kinase activity after chromatography or electrophoresis.

1. *Coupled Enzymic Assays*

Several coupled enzymic procedures have been developed in which the reaction rate is followed spectrophotometrically by measuring the appearance or disappearance of NADH or NADPH. These procedures have the advantage of being dynamic assays that enable continuous measurements, considerable sensitivity, rapidity, and ease of operation. In addition, they avoid the use of radioisotopes. Of the several methods available, the pyruvate kinase-lactate dehydrogenase and the hexokinase-glucose-6-phosphate dehydrogenase systems have been most popular.

a. Pyruvate Kinase-Lactate Dehydrogenase Method. The pyruvate kinase-lactate dehydrogenase procedure (7) follows the formation of ADP from ATP or GDP from GTP in a coupled reaction system con-

TABLE III
MICHAELIS CONSTANTS OF NUCLEOSIDE DIPHOSPHOKINASES FROM VARIOUS SOURCES^a

Nucleotides	Source of enzyme												
	Human erythrocytes (Refs. 7, 8, 24) pH						Brewer's yeast (Ref. 16)	<i>B. subtilis</i> (Ref. 9)	Calf liver mitochondria (Ref. 17)	Calf thymus (Ref. 30)	Beef heart mitochondria (Ref. 15)	Rat liver (Ref. 22) pH	
	5.4	5.8 ^b	6.3	6.8	7.3	8.3						6.05	6.50
ATP	0.2 ^c —	0.12 0.4— 1.0 ^c	3.0 ^c —	0.25 ^c —	0.08 ^c 0.1 ^c	0.17 ^c —	0.31 —	0.15 0.22 ^c	1.8–3.5 ^d ; 2.0–8.0 ^e	—	1.4 ^c	0.40 ^c	0.12 ^c
CTP	—	—	—	—	—	—	—	0.28; 0.42 ^c	—	—	—	—	—
GTP	0.2 ^c	0.074; 0.15 ^c	0.10 ^c	0.05 ^c	0.16 ^c	0.08 ^c	—	0.059; 0.10 ^c	—	—	—	0.11	0.10
ITP	—	—	—	—	—	—	—	0.30	—	—	—	—	—
dTTP	—	—	—	—	—	—	—	0.11	—	—	—	—	—
UTP	—	1.4	—	—	—	—	0.25	0.13	—	—	—	0.50 ^c	0.13 ^c

dCTP	—	—	—	—	—	—	—	—	—	—	—	—	—
dGTP	—	—	—	—	—	—	—	0.095	—	—	—	—	—
5F-UTP	—	1.4	—	—	—	—	—	—	—	—	—	—	—
ADP	—	0.04	—	—	—	—	0.05	0.056 ^c	0.05–1.5 ^d 0.01–0.17 ^e	—	0.1 ^c	0.05 ^c	0.015 ^c
CDP	0.21 ^c	1.0 ^c	0.18 ^c	1.10 ^c	0.25 ^c	0.50 ^c	—	—	—	0.5	—	—	—
GDP	—	—	—	—	—	—	—	—	—	—	—	—	—
UDP	—	—	—	—	—	—	0.043	—	—	—	—	—	—
IDP	—	—	—	—	—	—	—	—	—	—	—	—	—
dCDP	—	5.0	—	—	—	—	—	—	—	0.56	—	—	—
dGDP	—	0.13	—	—	—	—	—	—	—	—	—	—	—
dTDP	0.11 ^c	0.26; 0.55 ^c	0.22 ^c	0.20 ^c	0.30 ^c	0.12 ^c	—	—	—	0.39	—	0.40 ^c	0.50 ^c
8-azaGDP	—	0.1	—	—	—	—	—	—	—	—	—	—	—
6-azaUDP	—	3.3	—	—	—	—	—	—	—	—	—	—	—

^a All K_m values are expressed in mM and are apparent values unless otherwise mentioned.

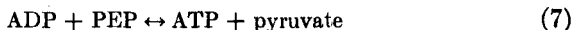
^b The enzyme was mentioned as homogeneous NDP kinase (7) but was found later to be a mixture of pI 5.8 and 6.3 with pI 5.8 as a main component (8).

^c Corrected K_m (apparent K_m value increases as the second substrate approaches saturation concentration).

^d In presence of a fixed amount of Mg^{2+} .

^e In presence of an excess of Mg^{2+} .

taining phosphoenolpyruvate (PEP), NADH, pyruvate kinase, lactate dehydrogenase, and the nucleotide substrates plus the appropriate salts and buffers according to the following reactions:



The rate of decrease in absorbance at 340 nm is followed spectrophotometrically. Although this procedure is very convenient, it has certain limitations. The diphosphate nucleotide substrate must be one that is a very poor substrate for pyruvate kinase, and the triphosphate nucleotide substrate must liberate a diphosphate nucleotide that is a good substrate for pyruvate kinase. This restricts somewhat the nucleotide combinations that may be studied. Diphosphate nucleotides such as dGDP, 8-azaGDP, dTDP, dCDP, and dADP fulfill these conditions (7, 61). Of the diphosphate nucleotides readily available, dTDP in concentrations as great as 1 mM gives negligible background velocities with the pyruvate kinase indicator system (8).

b. Hexokinase-Glucose-6-Phosphate Dehydrogenase Assay. The hexokinase-glucose-6-phosphate dehydrogenase method was developed originally by Berg and Joklik (3) who measured the formation of ATP from ADP and ITP or UTP in the presence of hexokinase-glucose-6-phosphate dehydrogenase indicator system according to the following equations:



Although this method is convenient, the only diphosphate nucleotide that can be studied is ADP and triphosphate nucleotide must be a poor substrate for hexokinase. Various descriptions of this method have been given (7, 51).

c. Other Coupled Enzymic Assays. Other coupled enzymic assays for NDP kinase have been developed, but their use has been limited to a few publications; for example, Sanadi *et al.* (29) have coupled the NDP kinase reaction with an α -ketoglutarate-succinate thiokinase system. Ratliff *et al.* (5) coupled the NDP kinase reaction to the phosphoglycerate kinase and triosephosphate dehydrogenase reactions, whereas Berg and Joklik coupled the NDP kinase reaction to the adenylate kinase

and adenylate deaminase reactions (31). Nakamura and Sugino (30) have developed a procedure that employs $\beta^{32}\text{P}$ -labeled dTDP. Upon completion of the NDP kinase reaction, the products are digested with a snake venom preparation that liberates both pyrophosphate and phosphate which are separated by a specific method. The reaction is followed by measuring the incorporation of ^{32}P into the pyrophosphate fraction (62).

2. Isotopic Procedures

One of the most commonly employed procedures for NDP kinase assays is the ATP-ADP exchange or a related isotopic method (14, 28, 63). In a typical procedure, a ^{14}C -labeled disphosphate nucleotide is incubated with a nonlabeled triphosphate nucleotide. Upon completion of the reaction, the di- and triphosphate nucleotides are separated, usually by chromatography, and the extent of incorporation of radioactivity into the triphosphate nucleotide is determined (8, 44, 52). Some laboratories have employed ^{32}P -labeled nucleotides for following the reaction (9, 15, 59, 63). The isotopic method has the advantage that it does not impose the restrictions of a coupled enzymic assay procedure and makes possible measurement of the reactions of any nucleotide pair depending on the availability of labeled substrates. This procedure has the disadvantage of being somewhat cumbersome and time-consuming and of employing radioisotopes. In comparison with the coupled-enzymic methods, isotopic methods are inconvenient for some kinetic analyses such as initial velocity studies. A number of isotopic methods have been described (3, 8, 9, 15, 16, 19, 28, 44, 52, 53, 63).

3. Staining Procedure for NDP Kinase

A convenient staining procedure has been developed for the identification of NDP kinase activity bands following gel electrophoresis or chromatography. This method is similar to the hexokinase-glucose-6-phosphate dehydrogenase procedure described above with the exception that the NADPH formed is detected by reaction with tetrazolium dyes (24). A useful method has been described where the complete indicator system is incorporated in an agarose overlay which becomes stained during the NDP kinase reaction (24). Since this staining method usually employs UTP and ADP as substrates, it is necessary to run control reactions for contaminating adenylate kinase (by omitting UTP).

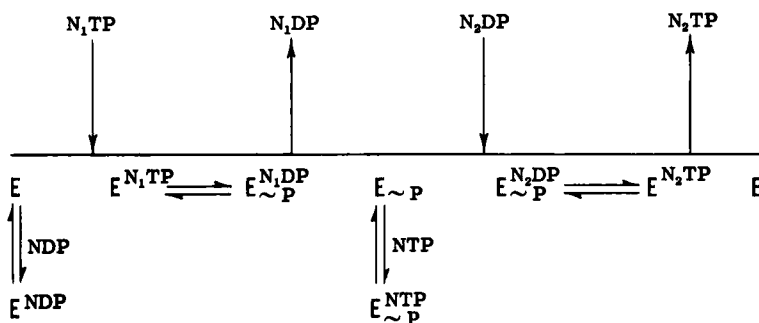
62. Y. Sugino and Y. Miyoshi, *JBC* **239**, 2360 (1964).

63. M. Chiga and G. W. E. Plaut, *JBC* **234**, 3059 (1959).

D. KINETICS AND CATALYTIC MECHANISM

1. Effect of Substrate Concentration

There have been a number of reports describing the kinetic behavior of NDP kinase isolated from erythrocytes (7, 8, 24), yeast (16), calf liver mitochondria (17, 28, 64, 65), calf thymus (30), beef heart mitochondria (15), and *Bacillus subtilis* (9). All enzymes that have been described to date display the classic kinetic behavior of a ping-pong reaction sequence (66) according to Scheme 1. With several NDP kinases,



SCHEME 1

including six isozymes from human erythrocytes, marked differences were observed in the kinetic parameters with a variety of nucleotide substrates as seen in Table III. In each case, the reactions were demonstrated to function via phosphorylated enzyme intermediates. Figure 2 shows the pattern of parallel lines observed with a human erythrocytic NDP kinase when the dGDP concentration was varied at several concentrations of GTP (7). In comparable experiments with other NDP kinase preparations, similar patterns of parallel lines have been described (9, 15-18). Figure 3 demonstrates the occurrence of the abortive complex, E-NDP, at high concentrations of a diphosphate nucleotide such as ADP. Here, the inhibition is partially overcome by increasing the concentration of the triphosphate nucleotide, ATP. Similar results have been obtained with other NDP kinase preparations (9, 15-17, 64). In studies of competing substrates, both di- and triphosphate nucleotides

64. A. Goffeau, J. Brachet, P. L. Pedersen, and A. L. Lehninger, *Arch. Int. Physiol. Biochim.* **76**, 179 (1968).

65. P. L. Pedersen, A. Goffeau, and A. L. Lehninger, *Fed. Proc., Fed. Amer. Soc. Exp. Biol.* **26**, 609 (1967).

66. W. W. Cleland, *BBA* **67**, 104 (1963).

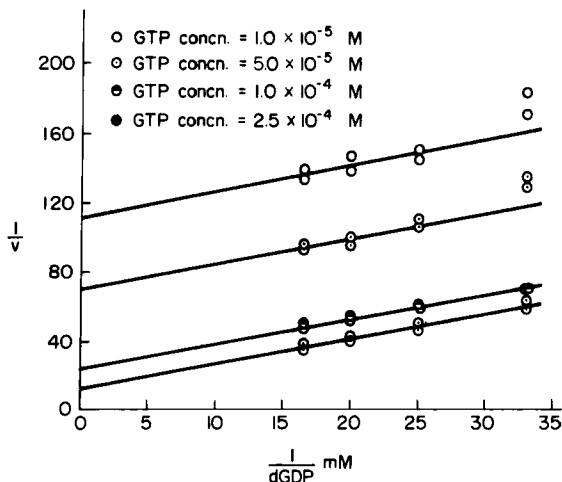


FIG. 2. Plots of reciprocal of the initial velocity ($v = -\Delta A_{340} \text{ min}^{-1}$) versus reciprocal of concentrations of dGDP (millimolar). Illustrates family of parallel line characteristics of the ping-pong reaction sequence. From Mourad and Parks (7).

have given the predicted classic competitive inhibition patterns (7, 9, 16). Studies of the inhibition of erythrocytic NDP kinase by the monophosphate nucleotide, GMP, gave K_i values of $4 \times 10^{-3} M$ when dTDP or dGDP was the variable substrate and $6.5 \times 10^{-4} M$ when ATP was the variable substrate. In both cases, noncompetitive patterns were ob-

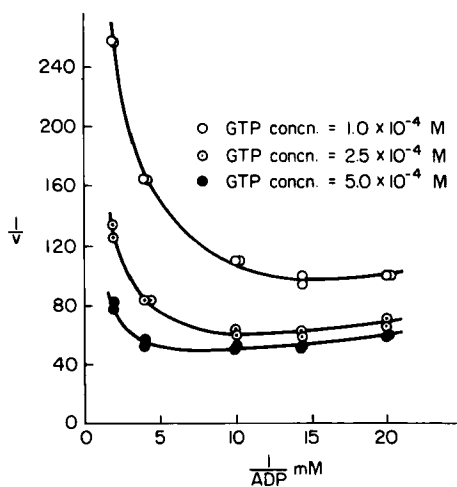


FIG. 3. Demonstration of abortive complex found at high concentration of ADP. See text for discussion. From Mourad and Parks (7).

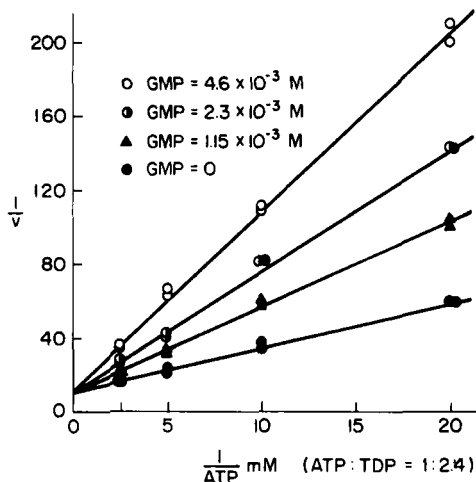


FIG. 4. Inhibition of erythrocytic NDP kinase by GMP with di- and triphosphate nucleotide substrates varied in a constant molar ratio. ATP and dTDP were varied in a ratio of 1:24. ATP and GMP were added in the concentrations indicated. Lineweaver-Burk plot is typical for competitive inhibition. From Mourad and Parks (7).

tained. However, when di- and triphosphate nucleotide substrates ATP and dTDP were varied in a constant ratio at different concentrations of GMP, the reciprocal plot yielded a family of straight lines that met at the $1/v$ intercept, a pattern of competitive inhibition (Fig. 4) (7). These results indicate that the inhibitor, GMP, competes for the nucleotide substrate-binding sites of both the free and the phosphorylated enzyme.

A rate equation offered by Garces and Cleland (16) for the mechanism of Scheme 1 which predicts initial velocities in the absence of products but which allows for the formation of abortive complexes by both substrates is

$$v = \frac{VAB}{K_a B(1 + B/K_{ib}) + K_b A(1 + A/K_{ia}) + AB} \quad (12)$$

where K_a and K_b are Michaelis constants for N_1TP (A) and N_2DP (B), and K_{ia} and K_{ib} are dissociation constants for N_1TP and N_2DP from abortive complexes with phosphorylated enzymes and free enzyme, and V is the maximal velocity.

2. pH Optima and Effect of Temperature

Several laboratories have described the pH optima for the reactions of various preparations of NDP kinase. In general, the optima range from

pH 6–9, and in most cases between pH 6.5 and 8.0 (8, 15, 28, 30, 31, 49). A Dixon analysis (67) of the effect of pH on the kinetic parameters has been performed with the pI 7.3 isozyme from human erythrocytes (8). In experiments with dTDP as the varying substrate, little change was observed in the Michaelis constant between pH 5.5 and 9.0. The plot of $\log V_{\max}$ versus pH revealed a relatively broad optimum with downward breaks at pH values of about 6.3 and 9.2. The downward break at pH 6.3 is consistent with the ionization of a histidine residue at the active site. It is intriguing that Wålinder (20) has identified phosphorylated residue of histidine and lysine in alkaline hydrolysates of human erythrocytic NDP kinase phosphorylated by reaction with [γ - ^{32}P]ATP. Since several studies have indicated that sulfhydryl groups play an important role in the activity of this enzyme, it is of interest that with the pI 7.3 isozyme a clearcut break is not seen in the plots of pK_m or $\log V_{\max}$ versus pH, from pH 8 to 9, which would be consistent with the occurrence of a sulfhydryl group (8).

Detailed studies of the temperature dependence of NDP kinase have been performed with the six isozymes from human erythrocytes and with a homogeneous preparation from *B. subtilis* (8, 9, 24). With the *B. subtilis* enzyme and the pI 7.3 and 8.3 isozymes from human erythrocytes, biphasic Arrhenius plots were observed. In each case, lower activation energies were observed at the higher temperatures. With the human erythrocytic isozymes, the break points occurred at about 31°, with the *B. subtilis* enzyme at about 25°. It is of interest that the other four human erythrocytic NDP kinase isozymes yielded linear Arrhenius plots. The fact that Arrhenius plots of two of the human erythrocytic isozymes are biphasic and the remaining four are linear has not yet been explained but suggests the existence of a fundamental difference in the physicochemical behavior of these isozymes.

E. METAL REQUIREMENTS

Many reports have appeared describing the metal dependency of NDP kinases (8, 9, 15, 17, 28, 30, 31, 49, 52). In no case has a requirement for a monovalent cation been described. However, all NDP kinases have an absolute requirement for divalent cations. Where examined, Mg^{2+} and Mn^{2+} have similar activity (8, 28, 30, 49, 52). Some differences have been observed from one NDP kinase preparation to the next in the activity of other divalent cations such as Ca^{2+} , Co^{2+} , and Zn^{2+} (8, 9, 28, 30, 31, 49, 52). In studies with NDP kinase from beef heart mito-

chondria, Colomb *et al.* (15) uncovered evidence indicating that the true substrates for the enzyme are Mg^{2+} -nucleotide complexes, such as Mg-ATP. In studies with a purified NDP kinase from liver mitochondria, it was found that Mg-ADP decreased the sensitivity of the enzyme to sulfhydryl reagents, whereas in the presence of Mg^{2+} alone the inactivation of the enzyme was enhanced. It was suggested that the Mg^{2+} stabilizes one of the equilibrium forms of the enzyme that possesses an essential sulfhydryl group that is more sensitive to reaction with mercurial agents (68).

F. SULFHYDRYL GROUPS

All NDP kinase preparations that have been examined give evidence of essential sulfhydryl groups (8, 14, 15, 28, 49, 57, 64, 68). In most cases, the enzymes are inhibited by mercurial reagents such as *p*-chloromercuribenzoate but they may be reactivated by the addition of a thiol such as cysteine, dithiothreitol, or mercaptoethanol (8, 14, 15, 68). With a homogeneous preparation of an NDP kinase from bovine liver mitochondria, titration studies revealed two classes of sulfhydryl groups. Four sulfhydryls per mole of enzyme are apparently free and react readily with *p*-chloromercuribenzoate, whereas a second group of four "hidden" sulfhydryl groups react with *p*-chloromercuribenzoate only when the enzyme is denatured with sodium dodecyl sulfate. Silver nitrate titrates both classes of sulfhydryl groups readily. Presumably the smaller size of the silver ion (Ag^+) allows it to reach the "hidden" sulfhydryls that are inaccessible to the much larger *p*-chloromercuribenzoate (57). It is also of interest that at low concentrations of a thiol reagent (below $10^{-7} M$) a slight stimulation of enzymic activity was observed (68).

Studies of the protective action of nucleotide substrates against the effects of thiol reagents have been performed with the human erythrocytic enzyme and bovine liver mitochondrial enzyme (8, 14, 68). In all cases, the presence of di- or triphosphate nucleotide substrates gave significant protection against inactivation by mercurials. Of particular interest is the observation that somewhat greater protection of the pI 7.3 isozyme from inactivation by *p*-chloromercuribenzoate was afforded by ATP or dTDP in the absence of a divalent cation (8). These observations suggest the occurrence of a nucleotide binding site on the enzyme that is not identical with the catalytic site but which influences the conformation of the enzyme. In other studies with the pI 7.3 human erythrocytic isozyme, the sensitivity to thiol reagents were increased by the addition

of 2 *M* urea. Moreover, in the presence of urea the enzyme could not be reactivated by addition of dithiothreitol which readily occurs in the absence of urea (8). This suggests the occurrence in the erythrocytic enzyme of at least two types of sulfhydryls as was observed with the bovine liver mitochondrial enzyme (57).

G. CONFORMATIONAL CHANGES

Several pieces of evidence indicate that NDP kinases from a variety of sources are susceptible to conformational effects. Where studied, the enzyme has proved to be polymeric and in the case of yeast NDP kinase the enzyme is a homologous hexamer. This suggests the possibility of interactions between subunits. With a homogeneous NDP kinase isolated from calf liver, a sigmoid-shaped rather than hyperbolic substrate saturation curve was obtained with ATP. This yielded a Hill coefficient of 1.8 (17).

Conformational changes have also been observed with the pI 7.3 isozyme from human erythrocytes. As noted above, marked protection against the inactivation by mercurial reagents is brought about by the addition of nucleotide substrates. Furthermore, the greatly enhanced susceptibility to mercurial reagents in the presence of 2 *M* urea is significantly diminished in the presence of a nucleotide substrate such as ATP (see Section III,F). Also, as noted above, with several NDP kinases diphasic downward Arrhenius plots are obtained which might indicate the existence of different conformational states as the temperature is varied (8, 9, 24).

IV. Functions in the Cell

It is generally assumed that NDP kinase is a major component of the enzymic pathway for the synthesis of triphosphate nucleotides, many coenzymes, and RNA and DNA. The activity of NDP kinases in many tissues is relatively high, usually 10- to 100-fold greater than the activity of the monophosphate nucleotide kinases. Furthermore, the NDP kinases studied to date do not display absolute specificity for nucleotide substrates, which is in sharp contrast to the monophosphate nucleotide kinases which are highly substrate-specific. Therefore, it seems unlikely that major control points in the pathways of triphosphate nucleotide synthesis occur at the level of NDP kinases and perhaps would be found at the level of the monophosphate nucleotide

kinases or at an earlier step. However, there are several indications that NDP kinases are responsive to the energy charge of the cell (69) and show conformational changes (8, 17) that are often seen in regulatory enzymes. In view of the high activity of NDP kinases in many cells, one would expect to observe a very rapid, indeed, almost instantaneous, equilibration of the high-energy phosphate bonds among the various nucleotide species. This should result in the intracellular ATP:ADP ratio being reflected in the triphosphate:diphosphate nucleotide ratios in the guanine, uracil, etc., polyphosphate nucleotide series. However, where nucleotide pools have been examined in cells that contain high levels of NDP kinase, such corresponding polyphosphate nucleotide ratios have not always been seen (33). This may, in some instances, result from compartmentation in the cell.

In the consideration of presumed NDP kinases, one must continually bear in mind the possibility that an enzyme that has low specificity for di- and triphosphate nucleotide substrates and that functions through the reversible formation of a high-energy phosphate-to-enzyme bond can behave as an NDP kinase. For example, an enzyme that functions via the following reaction sequence might be considered an NDP kinase if it has low specificity for nucleotide substrates and if one examines only reaction (13) of the sequence:



In this case, X and Y could be any non-nucleotide metabolites. Although to date there have been no definitive reports of this type of correlation between a presumed NDP kinase and other enzymic activities, the possible occurrence of such reactions must be borne in mind. For example, a succinic thiokinase that can react with adenine or guanine nucleotides (70) would behave as an NDP kinase in the presence of ATP and GDP. Mention should be made of the recent report of NDP kinase activity in highly purified preparations of DNA polymerase from bacterial sources (47). It is conceivable that the NDP kinase activity with this enzyme represents only a segment of a total reaction sequence. An observation that suggests the occurrence of specialized functions for NDP kinases is the wide variation in the kinetic parameters of the human erythrocytic isozymes with various nucleotide substrates (24). A finding that points to another function for an NDP kinase is the demonstration of an acid-stable, alkali-labile high-energy phos-

69. F. M. Thompson and D. E. Atkinson, *BBRC* **45**, 1581 (1971).

70. B. F. Burnham, *Acta Chem. Scand.* **17**, S123 (1963).

phate-enzyme bond in an NDP kinase from pig kidney (35), whereas most other NDP kinases form acid-labile, alkali-stable enzyme-phosphate bonds.

There is still much uncertainty about the role played by NDP kinases in such functions as mitochondrial oxidative phosphorylation, membrane transport involving ATPases, the substrate-level phosphorylation of GDP by the succinic thiokinase reaction, and the transport of high-energy phosphate from mitochondria to cytoplasm.

Since NDP kinases can catalyze ATP-ADP exchanges, there is still some confusion involving certain unique ATP-ADP exchange reactions found in mitochondria and NDP kinase reactions. These ATP-ADP exchange reactions are highly specific for adenine nucleotides (71). They are inhibited by both oligomycin and 2,4-dinitrophenol and are membrane-associated (39, 40), whereas NDP kinases are insensitive to these reagents, are nonspecific, and are not membrane-bound. An Mg^{2+} -dependent mitochondrial NDP kinase was separated from an oligomycin, 2,4-dinitrophenol-sensitive phosphorylating membrane fraction (41, 43). In digitonin-treated mitochondria, NDP kinase is associated with the outer membrane and the soluble fraction, whereas adenylate kinase occurs entirely in the soluble fraction (41, 43). In view of the recent finding of marked heterogeneity of NDP kinases in many tissues and subcellular fractions, including mitochondria (22), it is obvious that considerably more investigation will be required before definitive functions can be assigned to NDP kinases with confidence.

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